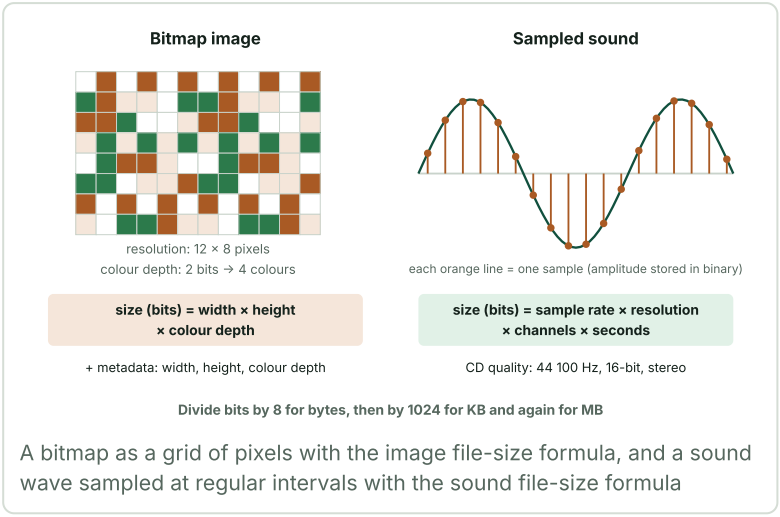
Representing images and sound

FD05
FACT FILE

Specification: students should be able to

- Describe how bitmap images are represented using pixels, resolution, colour depth and the RGB colour model
- Explain the purpose of metadata stored with an image file
- Compare bitmap and vector graphics and select the appropriate type for a task
- Describe how sound is digitised by sampling, including sampling rate and sample resolution
- Calculate the file sizes of uncompressed images and sound files and convert between units

Key words: pixel, bitmap, resolution, dots per inch (dpi), colour depth, RGB, metadata, vector graphic, pixelation, analogue, sampling, sampling rate, sample resolution (bit depth), analogue-to-digital converter (ADC), channel, Nyquist theorem, file size



A bitmap as a grid of pixels with the image file-size formula, and a sound wave sampled at regular intervals with the sound file-size formula

Digitising images and sound: an overview

Light and sound in the real world are analogue: they vary smoothly and continuously. A computer can only store discrete binary values, so images and sound must be digitised by measuring them and storing each measurement as a number. A bitmap image is digitised in space: the picture is divided into a grid of pixels and the colour of each pixel is recorded using a fixed number of bits, called the colour depth. Sound is digitised in time: an analogue-to-digital converter measures the amplitude of the sound wave thousands of times every second, and each sample is stored using a fixed number of bits, called the sample resolution. In both cases the same trade-off appears. Taking more measurements, through a higher resolution or a higher sampling rate, or storing each measurement more precisely, through a greater colour depth or sample resolution, makes the digital version a closer match to the original, but it also increases the file size, the storage needed and the time taken to transmit the file. Because the uncompressed size can be calculated exactly from these settings, file size calculations are a regular feature of exam papers. Vector graphics take a different approach, describing shapes mathematically rather than recording individual pixels.

Pixels, resolution and colour depth

A bitmap image is a two-dimensional grid of pixels, each of which stores one colour as a binary value. The resolution is the number of pixels in the image, usually written as width by height, such as 1920 by 1080. For printed work, resolution is often given in dots per inch: screen graphics are typically around 72 dpi, desktop printing around 150 dpi and professional printing 300 dpi or more. To find the pixel dimensions of a printed image, multiply each measurement in inches by the dpi. Colour depth is the number of bits used for each pixel, and n bits allow 2 to the power n colours, so 8 bits give 256 colours and 24 bits give 16,777,216. Most screens build colours from red, green and blue light, and 24-bit colour uses one byte for each of these three components, with values from 0 to 255. Increasing the resolution adds more pixels and therefore more detail, while increasing the colour depth allows smoother, more realistic shading. Both increase the file size, because the total number of bits is the number of pixels multiplied by the bits per pixel. If a bitmap is enlarged beyond its resolution, the individual pixels become visible, an effect called pixelation.

Metadata

The pixel data on its own is just a long sequence of bits, so an image file also stores metadata, which is data about the data. The most important items are the width and height of the image in pixels and its colour depth; without these, software could not tell where one row of pixels ends and the next begins, or how many bits belong to each pixel, and the picture could not be reconstructed correctly. Other metadata can include the resolution in dpi, the file format, the type of compression used and, for some formats such as GIF, the colour palette. Digital cameras and smartphones add further information known as EXIF data, such as the date and time a photo was taken, the camera model, exposure settings and frequently the GPS coordinates of the location. This extra information is useful for organising and searching photo collections, but it has privacy implications: a photograph shared online could reveal where a person lives or where a child goes to school, which is why many social media platforms remove location data when an image is uploaded. Metadata increases the file size slightly, but exam calculations normally ignore it unless the question says otherwise.

Bitmap and vector graphics compared

A vector graphic is not stored as pixels at all. Instead, it is a list of drawing objects, such as lines, circles, rectangles, curves and text, each described by properties including coordinates, size, line width, line colour and fill colour. Whenever the image is displayed or printed, the software recalculates the shapes at the required size, so a vector can be scaled up indefinitely without losing sharpness. A bitmap, by contrast, has a fixed number of pixels and becomes blocky when enlarged. The size of a vector file depends on how many objects it contains, not on how large it is displayed, so a company logo stored as a vector is usually very small and can be used on a business card or the side of a lorry from the same file. Vectors are also easy to edit, because each object can be moved, resized or recoloured independently. However, they are not suited to photographs, where colour changes continuously from pixel to pixel and cannot be described efficiently as shapes. Bitmaps are therefore used for photographs, scanned documents and realistic artwork, while vectors are preferred for logos, signs, icons, fonts, maps, charts and diagrams. SVG is the common vector format on the web.

Sampling sound

Sound travels as a pressure wave, and a microphone converts it into an analogue electrical signal whose voltage varies continuously. An analogue-to-digital converter, or ADC, measures this signal at regular, equally spaced intervals; each measurement is a sample of the wave's amplitude and is stored as a binary number. The sampling rate is the number of samples taken each second, measured in hertz, and CD-quality audio uses 44,100 samples per second. A higher sampling rate captures the shape of the wave more closely and allows higher frequencies to be recorded. The Nyquist theorem states that the sampling rate must be at least twice the highest frequency that needs to be reproduced; because human hearing extends to about 20 kHz, a rate a little over 40 kHz is sufficient. The sample resolution, or bit depth, is the number of bits per sample. With 16 bits there are 65,536 possible amplitude levels, so each sample is rounded to the nearest level; more bits mean less rounding error and a more faithful recording. Stereo recordings store two channels, one for each speaker. On playback, a digital-to-analogue converter turns the stored samples back into a varying voltage.

Calculating file sizes

Uncompressed file sizes follow directly from the way the data is represented. For a bitmap image, the size in bits is the width in pixels multiplied by the height in pixels multiplied by the colour depth. If dimensions are given in inches, first multiply each by the dpi to find the pixel width and height. For sound, the size in bits is the sampling rate multiplied by the sample resolution, the number of channels and the duration in seconds; remember to convert minutes into seconds first. Once you have the number of bits, divide by 8 to get bytes, by 1024 to get kilobytes, and by 1024 again for megabytes. Some questions, and some textbooks, use 1000 instead of 1024; the convention used by the question should be followed, and if none is stated, showing which you have used gains credit. Calculations can also be reversed: for example, the number of photos that fit on a memory card is the card's capacity in bytes divided by the size of one photo, rounded down because a partial photo cannot be stored. These are the sizes of uncompressed formats such as BMP and WAV; compressed formats such as JPEG and MP3 are far smaller because they discard or encode data more efficiently.

T Key terms from the mark schemes

Pixel
Short for picture element: the smallest addressable element of a bitmap image, storing a single colour as a binary value.

Colour depth
The number of bits used to represent the colour of each pixel; n bits give 2^n possible colours, and more bits give a larger file.
Credited: Bit depth

Bitmap
An image stored as a grid of pixels, with the colour of each pixel held as a binary number.
Credited: Raster image

Sampling rate
The number of samples of an analogue sound wave taken per second, measured in hertz.
Credited: Sample rate / sampling frequency

Analogue-to-digital converter (ADC)
A device that measures an analogue signal at regular intervals and converts each measurement into a binary value.

Pixelation
The blocky appearance of a bitmap image when it is enlarged beyond its resolution so that individual pixels become visible.

Resolution
The number of pixels in an image, usually given as width × height in pixels, or as dots per inch (dpi) for printed images.
Credited: Pixel density / dpi

Metadata
Data about data, such as an image's width, height and colour depth, stored with the file so that software can interpret it correctly.

Vector graphic
An image stored as a set of objects described by mathematical properties such as coordinates, size and colour, so it can be scaled without loss of quality.

Sample resolution
The number of bits used to store each sound sample, which determines how precisely each amplitude is recorded; also called bit depth.
Credited: Bit depth

Nyquist theorem
The principle that the sampling rate must be at least twice the highest frequency in the signal for it to be reproduced accurately.

C# example

Checking file sizes in C#

```
long width = 1024, height = 768, depth = 24;
long imageBits = width * height * depth;
double imageMB = imageBits / 8.0 / 1024 / 1024;
Console.WriteLine("Image: " + imageMB + " MB");          // 2.25

long rate = 44100, res = 16, channels = 2, secs = 180;
long soundBits = rate * res * channels * secs;
double soundMB = soundBits / 8.0 / 1024 / 1024;
Console.WriteLine("Sound: " + Math.Round(soundMB, 1) + " MB");   // 30.3
```

Exam focus

Calculation questions typically carry two to four marks, with method marks for each stage, so always write the formula, substitute the values and show every unit conversion; a final answer on its own risks losing most of the marks if it is wrong. Check what unit the question asks for and whether it expects 1000 or 1024. Common mistakes are forgetting to divide bits by 8, forgetting to convert minutes to seconds, leaving out the number of channels for stereo, and confusing colour depth in bits with the number of colours. When asked how many colours a bit depth allows, show 2 to the power n. For 'explain the effect' questions, give a chain of reasoning: a greater colour depth means more bits per pixel, multiplied across every pixel, so a larger file but more colours and smoother shading. For bitmap versus vector, use paired comparison points covering how the image is stored, what happens when it is scaled, what file size depends on and what each is suited to, and apply them to the context, such as a logo printed at many sizes. Metadata answers should name specific items like width, height and colour depth and say why software needs them.

? Check yourself

1. State what is meant by the colour depth of an image. [1]
2. State how many different colours can be represented with a colour depth of 16 bits. [1]
3. Calculate the file size, in kilobytes, of an uncompressed 640 × 480 pixel image with a colour depth of 8 bits. Show your working. [3]

Answers: 1. The number of bits used to represent the colour of each pixel (1). 2. 2^{16} (1) = 65,536 colours. [1 mark for 65,536] 3. 640 × 480 = 307,200 pixels (1) / × 8 bits = 2,457,600 bits ÷ 8 = 307,200 bytes (1) / ÷ 1024 = 300 KB (1).